$\zeta(s)$ AS AN SO(4) PROJECTION TO THE COMPLEX PLANE: HARMONIC–Δ CLOSURE, QUATERNION AXIS READOUT, AND REGIONAL CLOSED FORMS

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ABSTRACT. **Position.** This note synthesises the machine-checked S^3 / Riemann story spine in the public HQIV Lean library [4]. It sits on the closure record [3] (harmonic divergence $\Rightarrow$ phase lift Δ, $\langle \mathfrak{so}(3), \Delta_4\rangle_{\mathrm{Lie}} = \mathfrak{so}(4)$), the octonion light-cone carrier [2], and the TUFT+SM synthesis [1]. It is *not* a phenomenology paper; it is a formal packaging of $\zeta(s)$ as a projection geometry.
Main geometric claim. The functional-equation pair $(\sigma, 1-\sigma)$ is rotated at 45° in a quaternionic two-plane. The diagonal slot is fixed at $1/\sqrt{2}$ (even / i carrier); the equator slot $(2\sigma-1)/\sqrt{2}$ is the odd sin–cos carrier. Classical $\zeta(s)$ is then a *regional* closed form on $\mathbb{C}$: Dirichlet / Euler on $\Re s > 1$, a sin–cos–Γ–π functional-equation assembly on the open strip, Bernoulli values at negative integers, and π-sector Bernoulli closed forms at even positives.
Projection trichotomy. At even integers $s = 2k$ the angle $\pi s/2 = k\pi$ kills the sine (j–k) slot ($\sin = 0$, $\cos = (-1)^k$): even ζ-values are pure π-powers. At odd integers the cosine slot vanishes and sine carries the residue (no elementary closed form for $\zeta(3)$, $\zeta(5)$, $\ldots$). On the fractional open strip both slots are nonzero: the rotation ratio $\tan(\pi s/2)$ is the projection readout.
Reading the "weird" closed form. The open-strip formula displays $\Gamma(1-s)$ poles and sin/cos zeros that look singular. In the HQIV packaging these are not pathologies: negative-even trivial zeros are discharged from the nontrivial-zero slot; Γ poles are the even-sector continuation markers; together they force the *meaningful* zero-locus question onto the real critical line $\Re s = \frac{1}{2}$—the only place the equator projection $(2\sigma-1)/\sqrt{2}$ vanishes pointwise.
Status. Every algebraic / Mathlib link in the harmonic–prime–ζ chain and every 45° projection identity cited here is sorry-free in `lake build HQIVStory`. Unconditional *Riemann Hypothesis* is *not* claimed: the capstone predicate `InteriorAssemblyNonzeroAtNontrivialZerosOffLine` remains the single honest analytic gap (Appendix A).

Patch QFT.. The quantum field theory used here is the HQIV *patch QFT*: fields, actions, charges, amplitudes, and readouts live on finite horizon patches labelled by shell indices $m \in \mathbb{N}$, finite spacetime index types such as Fin 4, and discrete carrier charts. Each patch is the *entire accessible universe* in theory terms—the full causally accessible patch net, not a mesh tile waiting for a fundamental continuum limit. This is the operative QFT of the model. Its empirical content is the finite-patch calculation itself: mass and coupling ladders, curvature ratios, shell thermodynamics, anomaly traces, branching ledgers, and rational witnesses are computed on the patch carrier and then compared with laboratory observables. Continuum notation, when used, is a dictionary for readers trained in smooth field theory; it is not the definition of the theory.
Gauge and topology on the patch carrier. The patch QFT has no hidden continuum gauge sector whose consistency must be assumed. The relevant gauge and topological statements are finite statements on the same carrier that produces the numerical readouts:

- **Anomaly cancellation.** For one Standard-Model generation (Q_L, u^c, d^c, L, e^c, ν^c), the patch trace identities give zero $U(1)_Y^3$, gravitational–$U(1)_Y$, $SU(3)_c^2 U(1)_Y$, $SU(2)_L^2 U(1)_Y$, and $SU(3)_c^3$ anomaly coefficients; the $SU(2)_L^3$ coefficient vanishes because the weak doublet slot is pseudoreal. Three generations are a finite Fin 3 label sum of the same cancelled one-generation block, not a fitted multiplicity.

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- **Topological sector.** The patch abelian gauge datum has a single topological sector. Instanton number, Pontryagin number, first Chern number, and $U(1)$ winding number vanish identically on patch data; the θ contribution is therefore θ-independent; and abelian commutator obstructions vanish.
- **Non-abelian charts and holonomy.** Light-cone $\mathfrak{so}(8)$ Lie closure, $\mathrm{G}_2 \cup \{\Delta\}$ data, admissible holonomy on integrable Hopf shells, cyclic plaquette flatness, and weak/colour commutator certificates are discharged on the octonion carrier (see the sector zeta and holonomy discharge bundle in the Lean catalog). Where the optional strong-colour certificate is invoked, the global Gell-Mann bracket law is proved on the finite $\mathfrak{su}(3)$ chart.

External continuum theorems not required. Several textbook continuum constructions are *not* load-bearing prerequisites because the patch carrier already fixes the traces, holonomy transport, topological charges, and sector zeta bodies used for predictions: BPST instanton classification on a separate smooth four-manifold, Nielsen-style principal-bundle completeness on an external C^∞ base, and path-integral measure choice on that base. HQIV does not claim to re-prove those external theorems; it proves the finite patch statements that supply the same consistency checks for the readout pipeline. This is why familiar QFT language can appear in the comparison layer while the predictions remain patch-level statements with fewer adjustable inputs than a continuum effective fit.

1. Introduction

The Riemann zeta function is usually introduced as a Dirichlet series and then analytically continued. In the HQIV S^3 story the same object is instead a *projection readout*: the functional equation is a reflection symmetry $\sigma \leftrightarrow 1-\sigma$, the 45° twiddle separates a fixed diagonal channel from a free equator channel, and the divergent harmonic backbone of the closure paper forces a nonzero phase-lift connector Δ that closes $\mathfrak{so}(3)+\Delta$ to $\mathfrak{so}(4)$ on the four-dimensional quaternion carrier.

This note records what is already proved in Lean:

(i) the harmonic–Δ–SO(4) closure bridge;
(ii) the harmonic $\Rightarrow$ prime Euler product $\Rightarrow \zeta \Rightarrow \Lambda = -\zeta'/\zeta$ pipeline for $\Re s > 1$;
(iii) regional closed forms for $\zeta(s)$ matching Mathlib;
(iv) j/k axis rotation cancellation at even π-multiples and rotation ratios on odd / fractional slots;
(v) discharge lemmas excluding nontrivial zeros from half-planes and negative integer slots;
(vi) a factorization $\zeta(s) = h(s)\cdot(2\sigma-1)/\sqrt{2}$ on the open strip off $\sigma = \frac{1}{2}$, with $h =$ `interiorStripH`.

The remaining RH step is *not* continuation; it is proving that the assembly h does not vanish at nontrivial zeros off the line—equivalently, upgrading orbit cancellation (functional-equation pairs) to pointwise cancellation at actual ζ zeros.

2. Harmonic divergence forces Δ and SO(4)

The closure spine [3] proves a uniform lower bound

$$H_n := \sum_{m=0}^{n-1} \frac{1}{m+1} \;\le\; K(n) := \sum_{m=0}^{n-1} \underbrace{\frac{1}{m+1}\big(1+\alpha\log(m+1)\big)}_{\text{shell density}}, \qquad \alpha = \tfrac{3}{5},$$

with K strictly increasing (Hqiv/Story/S3ClosureDeltaLiftBridge.lean, Hqiv/Geometry/OctonionicLightCone.lean). The decomposition

$$K(n) = H_n + \alpha \sum_{m=0}^{n-1} \frac{\log(m+1)}{m+1}$$

is the exact curvature-integral split (`curvature\_integral\_eq\_harmonic\_plus\_alpha\_log`).

Δ is *not* identical to H_n, but it is forced because the normalized channel $\Omega = K/K_*$ diverges. On the toy four-dimensional carrier the Lie span of $\mathfrak{so}(3)$ together with the connector $\Delta_4 = J_{14}$ generates full $\mathfrak{so}(4)$ (`so3\_delta\_lifts\_to\_so4`).

2.1. **Third harmonic orbit and the $1 + \alpha/3$ projection.** The third partial sum $H_3 = 1 + \frac{1}{2} + \frac{1}{3}$ is the first shell carrying an explicit $1/3$ harmonic term. Against the HQIV imprint $\alpha = 3/5$ the projection multiplier

$$1 + \frac{\alpha}{3} = 1 + \frac{1}{5} = \frac{6}{5} = 1.2$$

is proved exactly (Hqiv/Story/S3HarmonicDeltaEvenOrbit.lean, `so4HarmonicThirdProjection\_eq\_one\_point\_two`). On the eight-dimensional shell the unit 7-sphere area proxy is $\pi^4/3$; halving at the 45° equator gives $\pi^4/6$ (`so4EquatorHalfArea\_eq\_pi\_four\_sixths`), parallel to the even Bernoulli slot $\zeta(2) = \pi^2/6$ at π^2 power.

3. Forty-five degree projection

Definition 3.1 (Functional-equation pair and 45° slots)**.** *For $\sigma \in \mathbb{R}$ define the pair $(\sigma, 1-\sigma)$ and the rotated coordinates*

$$\mathtt{rot45Diag}(\sigma, 1-\sigma) = \frac{\sigma + (1-\sigma)}{\sqrt{2}} = \frac{1}{\sqrt{2}},$$
$$\mathtt{rot45Free}(\sigma, 1-\sigma) = \frac{\sigma - (1-\sigma)}{\sqrt{2}} = \frac{2\sigma - 1}{\sqrt{2}}.$$

At $\theta = \pi/4$ one has $\cos\theta = \sin\theta = \sqrt{2}/2$ (not π); π enters through the functional-equation angle $\pi s/2$ below.

Theorem 3.2 (Critical line as equator)**.** *The free coordinate vanishes iff $\sigma = \frac{1}{2}$:*

$$\mathtt{rot45Free}(\sigma, 1-\sigma) = 0 \iff \sigma = \tfrac{1}{2}.$$

A nonzero twiddle shift $\pi/4 + \delta$ with $\sin\delta \neq 0$ moves the vanishing locus off $\sigma = \frac{1}{2}$ (Hqiv/Story/S3RotationRigidity.lean).

The complex lift of the equator readout is the **SO(4) critical factor**

$$\mathtt{so4CriticalFactor}(s) = \frac{2\sigma - 1}{\sqrt{2}}, \qquad s = \sigma + it,$$

which vanishes exactly on $\Re s = \frac{1}{2}$ (Hqiv/Story/S3ZetaClosedForm.lean).

4. Quaternion axis projection

On the unit imaginary quaternion shell $S^3 \subset \mathbb{H}$ the six axis poles $\pm i, \pm j, \pm k$ project to $\pm 1/\sqrt{2}$ on single axes and cancel as antipodal orbit pairs (Hqiv/Story/S3SixPolesResidual.lean, Hqiv/Story/S3PrimeAxisCancellation.lean).

Definition 4.1 (j/k slots in the FE channel)**.** *Define the sine and cosine projection slots*

$$\mathcal{S}(s) = \sin\frac{\pi s}{2}, \qquad \mathcal{C}(s) = \cos\frac{\pi s}{2}.$$

These are the j/k rotation content in the functional-equation odd channel; $\mathcal{C}$ is packaged as `zetaSinCosFactor` *in Lean.*

Theorem 4.2 (Even π-multiples: j/k cancellation)**.** *For $k \in \mathbb{N}$ at $s = 2k$:*

$$\mathcal{S}(2k) = \sin(k\pi) = 0, \qquad \mathcal{C}(2k) = \cos(k\pi) = (-1)^k.$$

Theorem 4.3 (Odd integers: cosine cancels)**.** *For $k \in \mathbb{N}$ at $s = 2k + 1$:*

$$\mathcal{C}(2k+1) = 0, \qquad \mathcal{S}(2k+1) = (-1)^k.$$

Theorem 4.4 (Fractional strip: rotation ratio)**.** *For $0 < \Re s < 1$ with $\mathcal{C}(s) \neq 0$,*

$$\frac{\mathcal{S}(s)}{\mathcal{C}(s)} = \tan\frac{\pi s}{2}.$$

At $\sigma = \frac{1}{2}$ both slots equal $\sqrt{2}/2$ and the ratio is 1 (`zetaAxisRotationRatio\_at\_half`*).*

Theorems 4.2–4.4 are proved in `Hqiv/Story/S3ZetaAxisRotationProjection.lean`. They explain why **even** ζ-values admit π-sector closed forms while **odd** positive integers (e.g. $\zeta(3)$) occupy the Apéry / plastic continuation slot with no elementary closed form in Mathlib.

5. Regional closed form for $\zeta(s)$

Table 1 collects the proved regional packaging. Every row is machine-checked against Mathlib in `Hqiv/Story/S3ZetaClosedForm.lean` and `Hqiv/Story/S3SO4ZetaProjectionClosedForm.lean`.

Table 1. Regional SO(4) channel packaging for $\zeta(s)$ (all rows proved).

Region	Channel	Closed form
$\Re s > 1$	even / Dirichlet	$\sum_{n\geq 0}(n+1)^{-s}$
$0 < \Re s < 1$	odd / sin–cos–Γ	FE assembly (`oddStripChannel`)
$s = 2k,\ k \geq 1$	even π-sector	$(-1)^{k+1}\frac{2^{2k-1}\pi^{2k}B_{2k}}{(2k)!}$
$s = -k$	Bernoulli	$-\frac{B'_{k+1}}{k+1}$
odd $s \geq 3$	odd continuation	no elementary closed form

5.1. Right half-plane.

Theorem 5.1 (Shell / Dirichlet sum)**.** *For $\Re s > 1$,*

$$\zeta(s) = \sum_{n=0}^{\infty}(n+1)^{-s},$$

the HQIV shell ladder with shift $(n+1)$ (`riemannZeta\_dirichlet\_closed\_form`*).*

This is the even harmonic–Euler sector: the harmonic–prime–ζ path proves the prime product and $\Lambda = -\zeta'/\zeta$ on the same region (`Hqiv/Story/S3HarmonicPrimeZetaPath.lean`).

5.2. Open critical strip.

Theorem 5.2 (Functional-equation closed form)**.** *For* $0 < \Re s < 1$*,*

$$\zeta(s) = 2\,(2\pi)^{-(1-s)}\,\Gamma(1-s)\,\cos\frac{\pi(1-s)}{2}\,\zeta(1-s). \tag{1}$$

Equation (1) is Mathlib's `riemannZeta_one_sub`, repackaged as `oddStripChannel` $= \zeta(s)$ on the strip (`Hqiv/Story/S3SO4InteriorWitness.lean`).

5.3. Even positive integers.

Theorem 5.3 (π-sector values)**.** *For* $k \geq 1$*,*

$$\zeta(2k) = (-1)^{k+1}\frac{2^{2k-1}\pi^{2k}B_{2k}}{(2k)!}.$$

In particular $\zeta(2) = \pi^2/6$ *and* $\zeta(4) = \pi^4/90$*.*

At these points $\mathcal{S}(2k) = 0$: the j/k rotation has completed full π-turns and only the even π-sector residue remains—consistent with Theorem 4.2.

5.4. Negative integers.

Theorem 5.4 (Bernoulli slot)**.** *For* $k \in \mathbb{N}$*,*

$$\zeta(-k) = -\frac{B'_{k+1}}{k+1}.$$

5.5. Master projection packaging.

Definition 5.5. *`zetaSO4Projected`$(s) := \zeta(s)$ with the regional characterisations of Table 1 as proved corollaries.*

Off the critical line on the open strip,

$$\zeta(s) = \texttt{so4EquatorProject}(s)\cdot\texttt{interiorStripH}(s), \qquad \texttt{so4EquatorProject}(s) = \frac{2\sigma-1}{\sqrt{2}},$$

when $0 < \sigma < 1$ and $\sigma \neq \frac{1}{2}$ (`zetaSO4Projected\_eq\_sin\_cos\_channel\_off\_line`).

6. Why the closed form looks weird—and why the answer is on the real line

The strip formula (1) exposes three features that appear singular when read naively:

(a) $\Gamma(1-s)$ has poles at $s = 1, 2, 3, \ldots$;
(b) $\cos(\pi(1-s)/2)$ has zeros at odd integers;
(c) negative-even *trivial* zeros of ζ sit at $s = -2, -4, \ldots$

The HQIV discharge chain treats these as **regional bookkeeping**, not as contradictions:

- **Negative-even trivial zeros** are the only negative-integer zeros; nontrivial zeros cannot occupy negative integer slots (`Hqiv/Story/S3FESlotDischarge.lean`, `nontrivial\_zero\_fe\_slot`).
- **Open strip** points avoid $\Re s > 1$ and $\Re s < 0$ by Mathlib nonvanishing and FE reflection (`Hqiv/Story/S3DeltaOrbitOffStrip.lean`).
- Γ **poles** on the strip are paired with cos zeros in the standard FE; they mark the even-sector continuation, not a complex off-line zero locus.

Remark 6.1 (Geometric reading)**.** *The "infinity" from Γ and the "zero" from negative-even slots are two faces of the same projection geometry: they clear the plane of spurious complex-axis candidates and leave the* pointwise *equator condition `rot45Free` $= 0 \Leftrightarrow \sigma = \frac{1}{2}$ as the only balanced imaginary readout (`Hqiv/Story/S3OrbitVsPointwiseGap.lean`). Orbit pairs $\{\sigma, 1-\sigma\}$ always cancel in sum; RH asks that actual ζ zeros satisfy the stronger pointwise condition—literally `AllNontrivialZerosOnLine` $\Leftrightarrow$ `RiemannHypothesis`.*

7. Explicit $h(s)$ on the strip

On the open strip with $\sigma \neq \frac{1}{2}$ the assembly collapses to the explicit quotient (Lean: Hqiv/Story/S3InteriorStripHClosedForm.lean):

$$h(s) = \frac{\zeta(s)}{\texttt{so4CriticalFactor}(s)} = \frac{\zeta(s)\sqrt{2}}{2\sigma - 1}. \tag{2}$$

Substituting the functional-equation assembly (1) before the quotient gives the full closed form with $\Gamma(1-s)$ and $\cos(\pi(1-s)/2)$ in the numerator (`interiorStripH\_fe\_closed\_form`): the "weird" regional singularities are then *visible in h itself*, not hidden inside ζ alone.

Theorem 7.1 (Reflection symmetry of the critical factor)**.** *For all $s \in \mathbb{C}$,*

$$\textit{so4CriticalFactor}(1-s) = -\textit{so4CriticalFactor}(s),$$

because the readout depends only on σ and swaps as $\sigma \leftrightarrow 1-\sigma$ (so4CriticalFactor_one_sub).

Proposition 7.2 (Reflection product seed (pathway A))**.** *For $0 < \sigma < 1$, $\sigma \neq \frac{1}{2}$,*

$$h(s)\,h(1-s) = -\frac{\zeta(s)\,\zeta(1-s)}{\textit{so4CriticalFactor}(s)^2}.$$

Multiplying h by the equator factor recovers ζ on the strip (`completedInteriorFromH\_eq\_zeta\_on\_strip`); the capstone predicate `InteriorStripHNonvanishingCapstone` is **logically equivalent** to `RiemannHypothesis` (`interior\_capstone\_iff\_RiemannHypothesis`)—not a weaker statement.

8. Interior factorization and RH status

On $0 < \Re s < 1$, $\sigma \neq \frac{1}{2}$, define

$$h(s) := \texttt{interiorStripH}(s) = \frac{\zeta(s)}{\texttt{so4CriticalFactor}(s)}.$$

Then $\zeta(s) = h(s) \cdot (2\sigma - 1)/\sqrt{2}$ off the line (Hqiv/Story/S3SO4InteriorWitness.lean).

Definition 8.1 (RH capstone predicate)**.** *InteriorAssemblyNonzeroAtNontrivialZerosOffLine h asserts: if $\zeta(s) = 0$ for a nontrivial zero with $\sigma \neq \frac{1}{2}$, then $h(s) \neq 0$.*

Theorem 8.2 (Conditional RH)**.** *If the capstone predicate holds for interiorStripH, then RiemannHypothesis follows (RiemannHypothesis_of_SO4_interior_witness).*

Remark 8.3 (Honesty)**.** *The factorization identity is proved; the capstone is* not*. Global factorization on the entire strip including $\sigma = \frac{1}{2}$ is false because $\zeta(\frac{1}{2} + it)$ is generally nonzero while the critical factor vanishes. The quotient witness $h = \zeta/$so4CriticalFactor proves algebra on off-line points only; it does not by itself discharge the capstone.*

The even channel `evenStripChannel` is presently degenerate on the strip because the odd FE channel already carries the full ζ; the physical even content is the harmonic–Δ multiplier $1 + \alpha/3$ and the $\pi^4/3$ sphere orbit (Hqiv/Story/S3HarmonicDeltaEvenOrbit.lean). A non-degenerate even/odd split is the remaining analytic packaging target.

9. Pathways toward an unconditional proof

The analysis below matches the claim-status table: the capstone is RH-hard. Plausible extensions that exploit *this* framework (not imported from external regularization claims) include:

- **A. Completed h.** Use (2) and $\texttt{so4CriticalFactor}(1-s) = -\,\texttt{so4CriticalFactor}(s)$ to build a candidate entire/completed interior function; check whether reflection forces off-line zeros to contradict discharge lemmas.

B. Zero-free re-expression. Rewrite h as a manifestly nonzero Dirichlet/Euler or harmonic-weighted integral on $\sigma \neq \frac{1}{2}$.

C. SO(4)/patch obstruction. Relate an off-line zero to a nonzero holonomy or commutator on the octonion carrier, contradicting proved $\mathfrak{so}(8)$ closure.

D. Classical tools in channel language. Phragmén–Lindelöf on vertical lines $\sigma \neq \frac{1}{2}$ using the $\tan(\pi s/2)$ ratio; explicit-formula with separated sin/cos slots.

E. Non-degenerate even split. Promote the harmonic–Δ multiplier into `evenStripChannel` so h is a genuine even/odd assembly rather than a pure quotient.

10. Lean module index

Build target: `lake build HQIVStory`. Core modules (import order in `HQIVStory.lean`):

- `Hqiv/Story/S3ClosureDeltaLiftBridge.lean` — harmonic $H_n \leq K(n)$, SO(4) lift;
- `Hqiv/Story/S3HarmonicPrimeZetaPath.lean` — harmonic $\Rightarrow$ primes $\Rightarrow \zeta \Rightarrow \Lambda$;
- `Hqiv/Story/S3FortyFiveProjection.lean` — 45° diagonal / equator;
- `Hqiv/Story/S3RotationRigidity.lean` — projection line, twiddle rigidity;
- `Hqiv/Story/S3ZetaClosedForm.lean` — regional closed forms, factorization predicate;
- `Hqiv/Story/S3SO4ZetaProjectionClosedForm.lean` — $\pi^4/6$ sphere slot, master packaging;
- `Hqiv/Story/S3HarmonicDeltaEvenOrbit.lean` — $1 + \alpha/3 = 6/5$ orbit;
- `Hqiv/Story/S3ZetaAxisRotationProjection.lean` — j/k cancellation and ratios;
- `Hqiv/Story/S3SO4InteriorWitness.lean` — even/odd channels, `interiorStripH`;
- `Hqiv/Story/S3FESlotDischarge.lean` — negative-integer slot discharge;
- `Hqiv/Story/S3DeltaOrbitOffStrip.lean` — off-strip confinement;
- `Hqiv/Story/S3OrbitVsPointwiseGap.lean` — orbit vs pointwise RH gap;
- `Hqiv/Story/S3RHDischarge.lean` — discharge packaging;
- `Hqiv/Story/S3InteriorStripHClosedForm.lean` — explicit $h(s)$, reflection symmetry, capstone $\Leftrightarrow$ RH.

Appendix A. Claim status table

Claim	Lean	RH-hard
$H_n \leq K(n)$, K strictly increasing	yes	no
$\langle \mathfrak{so}(3), \Delta_4 \rangle = \mathfrak{so}(4)$	yes	no
Shell sum $= \zeta(s)$ for $\Re s > 1$	yes	no
Prime Euler product $= \zeta(s)$	yes	no
FE strip closed form (1)	yes	no
Bernoulli / even-π regional values	yes	no
Even $s = 2k$: $\mathcal{S}(s) = 0$, $\mathcal{C}(s) = (-1)^k$	yes	no
Odd $s = 2k+1$: $\mathcal{C}(s) = 0$	yes	no
$1 + \alpha/3 = 6/5$ harmonic-third projection	yes	no
Nontrivial zeros $\notin$ negative integers	yes	no
Nontrivial zeros in open strip $0 < \Re s < 1$	yes	no
$\zeta = h \cdot (2\sigma - 1)/\sqrt{2}$ off $\sigma = \frac{1}{2}$	yes	no
`interiorStripH` $\neq 0$ at nontrivial zeros off line	no	yes
Unconditional `RiemannHypothesis`	no	yes

Appendix B. Notation

α	HQIV imprint 3/5 (`alpha\_eq\_3\_5`)
H_n	harmonic partial sum (`harmonicPartialSum`)
$K(n)$	curvature integral (`curvature_integral`)
`so4CriticalFactor`(s)	$(2\sigma - 1)/\sqrt{2}$
`oddStripChannel`	FE assembly (1)
`interiorStripH`	$(\texttt{even} + \texttt{odd})/\texttt{so4CriticalFactor}$
$\mathcal{S}, \mathcal{C}$	$\sin(\pi s/2)$, $\cos(\pi s/2)$

References

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